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# Evaluation of the Brazilian Portuguese version of linguistic inquiry and word count 2015 (BP-LIWC2015)

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## Abstract
Text psycholinguistic features are a valuable source for various research topics since they are used to obtain psychological, social, and linguistic aspects from written texts using dictionary files. These files are structured in categories, which are defined as groups of dictionary words that tap a particular domain (e.g., negative emotion words). The Linguistic Inquiry Word Count (LIWC) is a vastly used and versatile computer-based language analysis tool designed for text psycholinguistic analysis. The most recent version of the default English dictionary is LIWC2015, as it was released with the 2015 version of the LIWC software. The literature has recently introduced the latest Brazilian Portuguese LIWC dictionary (BP-LIWC2015), developed with the same categories as the LIWC 2015 English dictionary. However, the literature has also reported the need to evaluate BP-LIWC2015. In this scenario, this work investigates three questions: (i) Since LIWC2015 shows consistent improvements over the English dictionary developed in 2007 (LIWC2007), does BP-LIWC2015 achieves better text classification results than the older Brazilian Portuguese dictionary (BP-LIWC2007)? (ii) What is the equivalence between BP-LIWC2015 and BP-LIWC2007 with LIWC2015? (iii) Are there significant differences between Brazilian Portuguese dictionaries? To answer these questions, we conducted text classification experiments with four datasets and seven classification algorithms to compare the two Brazilian Portuguese LIWC dictionaries reported in the literature (i.e., 2007 and 2015). Second, we used a bilingual Portuguese-English scientific news collection to analyze the correlation between LIWC2015 and Brazilian Portuguese LIWC dictionaries. The results indicate that BP-LIWC2015 outperforms the older version in Brazilian Portuguese text classification. Finally, we found a more significant correlation between BP-LIWC2015 and the original English dictionary than the older version.

**Keywords** LIWC · LIWC2015 · Brazilian portuguese · Sentiment analysis · Text analysis · Machine learning.

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## 1 Introduction

Emotion plays a fundamental role in the processes of perception, memory, attention, and rational decision-making. People can express their emotions with voice, body language, and facial expressions and register them in various ways, such as in music or audio files, paintings, photography, and text in social media platforms (Calvo & D'Mello, 2010). Social media provides an environment where users worldwide tend to express their emotions (Dandannavar et al., 2020). Media-based social interactions occur frequently throughout the day and typically in text format, stimulated by the widespread use of social media platforms (Aggarwal, 2011). The growing number of media-based social interactions comes from the increasing use of portable devices over the past years (Falaki et al., 2010).

Sentiment Analysis (SA) research can use the huge amount of textual content available on social media platforms to reveal significant information about users or their published texts (Becker & Tumitan, 2013). The Sentiment Analysis area aims to detect and represent human opinions, sentiments, emotions, attitudes, and appraisals expressed in written text (Liu, 2020). Although studies in this area usually adopt polarity detection (e.g., if a commentary or phrase is positive or negative), it is also possible to find studies identifying a broader set of emotions (Cambria et al., 2013).

The representation of a text as a frequency vector or another vector does not intend to structure the data, but to represent an aspect of the unstructured data such as the word frequency (Zhang et al., 2010; Liu, 2012). Since the most important indicator of sentiments is sentiment words (Liu, 2020), several approaches use dictionaries to process the texts and create word count vectors. These dictionaries distribute words into conceptual dimensions, called categories, assigning them to a limited number of features to reduce the complexity of natural language (Grimmer & Stewart, 2013). Under such conditions, these dictionaries can assist in revealing and obtaining insights from relevant aspects of text by exploring several categories. Examples include affective words, such as *positive emotion*, *negative emotion* (e.g., 'love', 'happy', 'hope'), and *function words* such as *pronouns*, *articles*, and *prepositions* (e.g., 'what', 'once', 'they') (Tausczik & Pennebaker, 2010; Yin et al., 2019).

The literature presents dictionaries designed to perform Sentiment Analysis (Liu, 2020) with only two categories, such as Polyglot (Al-Rfou et al., 2014) or AffectPT-br (Carvalho et al., 2018). Other dictionaries are more complex, such as the Linguistic Inquiry Word Count (LIWC) (Pennebaker et al., 2015), which categorizes items such as words, word stems, phrases, and emoticons in a set of linguistic and psycholinguistic features. Here it is important to note that written text's linguistic and psycholinguistic features are a valuable source for various research topics (Bryman & Bell, 2015; Santos et al., 2016; Hernández Farías et al., 2019). This assumption is justified since the analysis of emotional expressions reveals aspects of psychological processes (Calvo & D'Mello, 2010; Tausczik & Pennebaker, 2010).

LIWC is one of the vastly used and most powerful computer-based language analysis tools worldwide (Dudău & Sava, 2020) and has been reported in numerous studies. Some of them process texts to identify social networks user's characteristics, such as gender (Schler et al., 2006; Hernández Farías et al., 2019), and age (Hernández Farías et al., 2019), which may have as its final application the identification of

possible sexual predators (Rodrigues et al., 2017). Studies also encompass the use of LIWC to investigate the lyrical content of songs across conditions of social and economic threats (Pettijohn & Sacco, 2009), assert marketing tendencies (Tang & Guo, 2015), detect mental illness (Shibata et al., 2016), and identify deception (Fornaciari et al., 2020). Furthermore, research with LIWC can provide new insights about language use by currently-depressed, formerly-depressed, and never-depressed people (Rude et al., 2004).

There are different versions of LIWC dictionaries in the literature. The most recent version of the default English dictionary is LIWC2015, as it was released with the 2015 version of the LIWC software (Pennebaker et al., 2015). In addition, LIWC2015 provides a set of dictionaries adapted to other languages and available for download at http://www.liwc.net/dictionaries, including Dutch (Wissen & Boot, 2017), German (Meier et al., 2019), Romanian (Dudău & Sava, 2020), and Brazilian Portuguese (Carvalho et al., 2019). Unlike the Brazilian Portuguese LIWC dictionary's latest version (BP-LIWC2015), the previous version of the Brazilian Portuguese 2007 dictionary, i.e., the BP-LIWC2007 (Balage Filho et al., 2013), is not compatible with the broader range of categories presented in LIWC2015.

Although the 2015 Brazilian Portuguese version, BP-LIWC2015, is still recent, the literature reports studies using this version to perform their experiments (Läubli, 2020; Barbosa et al., 2021; Mello et al., 2021; Sender, Carvalho & Guedes, 2021). However, the current text analysis literature highlights the need for studies to evaluate BP-LIWC2015 (Dudău & Sava, 2020). This work's importance (and novelty) is to address this need. In this scenario, we investigate three questions: (i) Since LIWC2015 shows consistent improvements over LIWC2007 (Eichstaedt et al., 2021), does BP-LIWC2015 achieves better text classification results than BP-LIWC2007? (ii) What is the equivalence between BP-LIWC2015 and BP-LIWC2007 with LIWC2015? (iii) Are there significant differences between Brazilian Portuguese dictionaries?

To answer these questions, we conducted a set of experiments based on preliminary LIWC evaluation studies (Wissen & Boot, 2017, Meier et al., 2019, Dudău & Sava, 2020). First, we compared BP-LIWC2015 and BP-LIWC2007 performance in classification tasks with four social network datasets and seven classification algorithms. The objective is to evaluate the relevance of features extracted by LIWC2015 and LIWC2007 with machine learning algorithms since these algorithms depend on the quality of features (Bengio et al., 2013). In other words, more meaningful features result in greater accuracy in the classification task (Midhun et al., 2014).

Next, we evaluated the equivalence between BP-LIWC2015, BP-LIWC2007, and LIWC2015 since equivalence between different dictionaries is a measure of criterion validity. Mainly the goal is to evaluate the capacity of BP-LIWC2007 and LIWC2015 to capture the same trends as LIWC2015 within similar datasets. To conclude, we investigated the coverage of the three previously mentioned dictionaries. The coverage can assist in the interpretation of results.

The remainder of this work is structured as follows. Section 2 reviews related works regarding the evaluation of LIWC dictionaries. Section 3 describes the LIWC application and presents the distributions of items in the current and previous versions of the dictionary. Section 4 discusses the method and materials. Section 5 provides

the experiments to evaluate the 2015 Brazilian Portuguese LIWC dictionary categories. Finally, Sect. 6 summarizes the contributions and presents the conclusions.

## 2 Related work

This section presents research related to evaluating the LIWC dictionaries in other languages. It then describes preliminary experiments with the Brazilian Portuguese LIWC dictionaries. Finally, it reveals how this study is based on the discussed research.

The Spanish LIWC 2007 dictionary (SP-LIWC2007) was evaluated in terms of a classification task into five categories (i.e., positive, negative, neutral, highly positive, or highly negative) by using different machine learning algorithms (del Pilar Salas-Zárate et al., 2014). The authors conducted the experiments with two Spanish datasets. The first one was a movie review dataset, and the second one corresponded to technological product reviews. In the first step, the authors processed both datasets using LIWC to extract linguistic features; next, they applied three machine learning algorithms to perform the classification task.

After presenting an automated process to translate the English LIWC dictionary into Dutch (NL-LIWC2015), Wissen and Boot processed a set of parallel Dutch and English texts (Wissen & Boot, 2017). Then, they processed these texts using NL-LIWC2015 and LIWC2015 dictionaries and compared the results presenting the correlation coefficients for each category. They found poor correlations for the informal word categories but observed that emotion, social words, and cognitive words performed well.

Meier et al. described the 2015 version of the LIWC dictionary in German (DE-LIWC2015) (Meier et al., 2019). Along with details about the psychometric properties, they analyzed over 34,000 text files and compared DE-LIWC2015 with DE-LIWC2001 (the previous 2001 version of the German dictionary), reporting means and correlations between the two versions. They also used LIWC analysis of German-English parallel corpora and translated transcripts to compare DE-LIWC2015 with LIWC2015. Moreover, the authors reported mean values and Pearson correlation between the DE-LIWC2015 and the English LIWC2015.

Dudău and Sava introduced Ro-LIWC2015, the first Romanian version of LIWC2015 (Dudău & Sava, 2020). They applied methodologies analogous to the ones the authors mentioned above. The goal was to evaluate LIWC in other languages and discuss linguistic aspects of the Romanian dictionary. They first tested if the Romanian and the English dictionaries provided similar outcomes when analyzing a parallel corpus and then analyzed posts from help-seeking forums for anxiety, depression, and health issues to assess the criterion validity of Ro-LIWC2015.

The first published assessment for the Brazilian Portuguese dictionary (Balage Filho et al., 2013) used the affective categories (i.e., 'positive emotion' and 'negative emotion') to compare BP-LIWC2007 with other two Portuguese dictionaries: Opinion Lexicon (Souza et al., 2011) and SentiLex (Silva et al., 2012). The authors evaluated the equivalence of BP-LIWC2007 with Opinion Lexicon and SentiLex to provide an internal evaluation. The experiments were performed firstly by generating

word count vectors considering the three dictionaries and lastly by applying a sentiment classification task. The authors highlighted the poor results of other dictionaries in detecting negative valence in texts. It is noteworthy that the proposed study conducts the experiments using a single dataset composed of 2,056 reviews from 13 different books (approximately 200 reviews for each book).

In the initial evaluation of BP-LIWC2015 (Carvalho et al., 2019), the authors evaluated LIWC categories by comparing BP-LIWC2015 against BP-LIWC2007 results in classification tasks. The first experiment produced word count vectors with BP-LIWC2015 and BP-LIWC2007, considering a three-class age group dataset. Next, they applied five traditional machine learning algorithms to perform a classification task to predict the age classes. The authors also created word count vectors considering a two-class sentiment dataset for sentiment polarity classification. The experiments conducted with BP-LIWC2015 outperformed BP-LIWC2007 with all five algorithms.

In this scenario, this work brings experiments similar to those performed with SP-LIWC2007, Ro-LIWC2015, DE-LIWC2015, and NL-LIWC2015 to evaluate BP-LIWC2015. The first step is to evaluate if BP-LIWC2015 presents relevant results in the classification task. The evaluation considered four social network datasets to compare BP-LIWC2015 with BP-LIWC2007. It is important to note that, unlike the BP-LIWC2007 evaluation, the experiments in this work use all categories from BP-LIWC2007 and BP-LIWC2015 dictionaries, aside from using seven classification algorithms. Next, the present study investigates the equivalency of BP-LIWC2015 and LIWC2015 dictionaries using a Portuguese-English collection of texts. Finally, we investigate the coverage of BP-LIWC2015 and BP-LIWC2007.

## 3 LIWC

LIWC is a software that processes text files according to a dictionary with items such as words, word stems, phrases, and emoticons associated with a set of categories (Tausczik & Pennebaker, 2010). Initially, the idea of LIWC development was to identify the group of words most studied in social, health, and personality psychology (Pennebaker, 2013). Then, the goal was to investigate if the ways people wrote about their traumatic experiences could predict improvements in their health (Pennebaker, 2013). According to Pennebaker and his collaborators (2015), associating words in the LIWC categories required several stages of development over several years.

**Table 1** An example of an output file containing the percentage of words in main LIWC categories

| File Number | Function words | Verbs | Adjectives | Interrogatives | … | Informal Language |
|---|---|---|---|---|---|---|
| 1 | 35.57 | 21.95 | 9.62 | 6.50 | … | 0.02 |
| 2 | 29.6 | 16.7 | 18.30 | 11.30 | … | 0 |
| 3 | 16.66 | 31.33 | 3.33 | 0.7 | … | 0 |
| 4 | 21.73 | 20.8 | 13.04 | 8.69 | … | 0.57 |
| … | … | … | … | … | … | … |
| n | 14.28 | 14.28 | 5.07 | 2.7 | … | 0 |

The number of dictionary items and categories has expanded considerably since the first LIWC English dictionary development, and now, LIWC2015 has 6,400 items such as words, word stems, phrases, and emoticons assigned to 73 categories (Pennebaker et al., 2015). The main categories in the LIWC dictionary represent linguistic, psychological, and social processes, structured with multiple levels of subcategories. For example, the affective category is subdivided into two subcategories: Positive Emotions (abbreviated as *posemo*) and Negative Emotions (abbreviated as *negemo*). The *negemo* subcategory is further divided into three subcategories: anxiety (abbreviated as *anx*), anger (*anger*), and sadness (abbreviated as *sad*) (Tausczik & Pennebaker, 2010).

The LIWC program process text files using dictionary files and increases the category's percentage according to each item found (Pennebaker et al., 2015). After processing a set of text documents, the LIWC program outputs a structured data file containing the document identification and the percentage value for each category in that document. Then this data file can be used as input data for classification algorithms. Table 1 represents an example of LIWC's output file from processing a given number of *n* text documents.

The first column in Table 1 shows an identification number for each text file. The second column shows the percentage of words matching the *Function words* category. The following columns show the percentage of words found in each of the following categories (*Verbs*, *Adjectives*, *Interrogatives*, …, *Informal Language*). In short, each line represents one text file, where the columns represent the percentage of words assigned in the current category.

The items in BP-LIWC2015 are translated from LIWC2015 and acquired from several other sources, such as pronouns and netspeak lists (Carvalho et al., 2019). One can note a difference in the number of categories and dictionary items when comparing Brazilian Portuguese dictionaries (BP-LIWC2015 and BP-LIWC2007). This difference occurs because BP-LIWC2015 was based on LIWC2015 and BP-LIWC2007 was based on LIWC2007. BP-LIWC2007 contains 127,149 dictionary items arranged in 64 categories (Balage Filho et al., 2013). On the other hand, BP-LIWC2015 has 14,459 items such as words, word stems, phrases, and emoticons arranged in 73 categories (Carvalho et al., 2019). Table 2 summarizes the number of dictionary items in each main category in LIWC2015, BP-LIWC2007, and BP-LIWC2015.

The first column brings the categories' names. The third, fourth, and fifth columns refer to the number of dictionary items in each category of LIWC2015, BP-LIWC2007, and BP-LIWC2015, respectively. For short, each line represents the number of items such as words, word stems, phrases, and emoticons found in each category in these three dictionaries.

The much larger number of items in BP-LIWC2007 does not positively impact a text analysis task since this number is also a result of multiple items with the same stem before an asterisk (*), causing inconsistencies and lengthening the text analysis time (Carvalho et al., 2019). Previous research also indicates BP-LIWC2007 contains items with categories improperly assigned (Carvalho et al., 2018); for example, the personal pronouns 'ela' (she) and 'ele' (he) are assigned to the *Impersonal pronouns* category. Another relevant issue is related to spelling; BP-LIWC2007 includes

**Table 2** Number of dictionary items in main categories of LIWC2015, BP-LIWC2007, and BP-LIWC2015

| Category | LIWC2015 | BP-LIWC2007 | BP-LIWC2015 |
|---|---|---|---|
| Function words | 491 | 5,512 | 1,426 |
| Verbs | 1,000 | 23,873 | 6,162 |
| Adjectives | 764 | NA | 1,241 |
| Comparisons | 317 | NA | 300 |
| Interrogatives | 48 | NA | 23 |
| Numbers | 36 | 83 | 106 |
| Quantifiers | 77 | 622 | 160 |
| Affect | 1,393 | 28,475 | 2,105 |
| Social | 756 | 13,634 | 1,445 |
| Cognitive Processes | 797 | 46,308 | 2,691 |
| Perceptual Processes | 436 | 17,607 | 1,001 |
| Biological Processes | 748 | 17,861 | 1,924 |
| Drives | 1,103 | NA | 3,176 |
| Past Focus | 341 | 7,684 | 3,159 |
| Present Focus | 424 | 4,715 | 1,416 |
| Future Focus | 97 | 268 | 1,456 |
| Relativity | 974 | 24,966 | 3,220 |
| Work | 444 | 7,735 | 781 |
| Leisure | 296 | 6,331 | 556 |
| Home | 100 | 2,019 | 145 |
| Money | 226 | 5,353 | 356 |
| Religion | 174 | 2,066 | 343 |
| Death | 74 | 2,429 | 90 |
| Informal Language | 380 | NA | 419 |

unusual variations not seen even in informal writing (Carvalho et al., 2018). The BP-LIWC2007 contains non-existent number inflections (e.g., 'issos'), misspelled words (e.g., 'ninguén' instead of 'ninguém'), and improper accent marks and auxiliary writing signs (e.g., 'ércebido', 'oiçei').

The categories not found in BP-LIWC2007 were represented by 'NA' (Not Apply). When compared to BP-LIWC2007 categories with the same name, BP-LIWC2015 reduced the number of words significantly. Notwithstanding, initial evaluations indicate that BP-LIWC2015 outperforms BP- LIWC2007 in classification tasks (Carvalho et al., 2019).

## 4 Materials and methods

This study evaluates BP-LIWC2015 by comparing it with LIWC2015 and BP-LIWC2007. The comparison between BP-LIWC2015 and BP-LIWC2007 uses four datasets with textual content extracted from social networks, exclusively in Brazilian Portuguese. The evaluation of BP-LIWC2015 compared to LIWC2015 uses a Portuguese-English bilingual collection of texts. After processing the selected text

files with the LIWC program, the Weka (Hall et al., 2009) program is used to produce the experiments with machine learning algorithms.

The first dataset used in this study is named MQD90k. It consists of 90,000 entries extracted from https://www.meuqueridodiario.com.br (MQD), a Brazilian online social network platform similar to a personal diary. Users usually record their everyday experiences by writing in the Brazilian Portuguese language. The dataset is divided into three classes: *10s*, *20s*, and *30s*, containing users of both genders with ages between 13 and 17, 23 to 27, and 33 to 42 years, respectively. The 6-year interval between the end of an age range and the beginning of another was planned for a more precise differentiation as done in other research (Schler et al., 2006; Carvalho et al., 2019). The *10s*, the *20s*, and the *30s* classes have 55,369, 20,958, and 17,311 posts, respectively. This dataset is publicly available at https://eic.cefet-rj.br/~lacafe/mqd-age-90k/.

The second dataset is named MQD60k (Carvalho et al., 2018). It contains 59,166 posts extracted from MQD online social network. A total of 32,244 posts are associated with happiness class and 26,922 with sadness class. This dataset is openly available on https://eic.cefet-rj.br/~lacafe/mqd60k/.

The third dataset in this work is derived from the openly available dataset TAS-PT. It can be downloaded at https://github.com/pauloemmilio/dataset. It contains two files, one with IDs of positive sentiment tweets and the other with IDs of negative sentiment tweets. However, the TAS-PT dataset does not include the textual content of tweets. Thus, the Twitter API was used to retrieve the content of the messages through the IDs contained in the files, resulting in the TAS-PT-60k dataset, with 59,260 textual content of tweets from TAS-PT, where 28,853 are negative, and 30,407 are positive (Carvalho et al., 2018).

The fourth dataset is PTSA-800k, composed of approximately 800,000 Twitter posts with textual content. It is also publicly available on https://www.kaggle.com/datasets/augustop/portuguese-tweets-for-sentiment-analysis/download. The PTSA-800k dataset contains smaller subsets with 50,000, 100,000, 200,000, 300,000, 400,000, and 500,000 posts. The subset TW-PTSA-100k, with approximately 100,000 posts, was selected for this study. It contains 49,996 positive and 49,999 negative labeled tweets.

The Portuguese-English collection of texts is referred to as FAPESP-CORPUS in this study. It is a bilingual Portuguese-English corpus composed of online editions of the FAPESP RESEARCH journal (http://revistapesquisa.fapesp.br). It contains approximately 2,700 parallel documents. The corpus is available for download at http://www.nilc.icmc.usp.br/nilc/tools/fapesp-corpora.tar.gz (Aziz & Specia, 2011).

This study adopts WEKA software version 3.6.15 for all classification tasks, considering the default configurations of the adopted algorithms. The Naive Bayes (NB) and the Multinomial Naive Bayes (MNB) algorithms are adopted as they are basic reference methods for text classification (Wang & Manning, 2012). The NB algorithm works using Bayesian decision, defining the event probability through previous knowledge associated with the event (Langley et al., 1992). The MNB algorithm works with a variant of NB using a multinomial model to obtain the variable frequency into the dataset (McCallum & Nigam, 1998). We also include the less investigated Logistic Regression classification method (Kleinbaum et al., 2002) because the

literature indicates that it is still used in practical tasks and is one of the most accurate classification methods (Pranckevičius & Marcinkevičius, 2016).

The Support Vector Machine (SVM) algorithm was also selected as part of this study. SVM transforms the training data into a higher-dimensional space based on a kernel function and then finds support vectors to maximize the separation between the classes (Han et al., 2011). This algorithm has been proven to be one of the most accurate classification algorithms (Sebastiani, 2002; Lan et al., 2009). WEKA provides the Sequential Minimal Optimization (SMO) algorithm (Platt, 1998) as the SVM implementation. Worthy of mention, it does not require a quadratic programming solver, differing from most SVM algorithms (Flake & Lawrence, 2002).

The other three algorithms use decision tree methods, which are the Random Forest (RF) algorithm, the Logistic Model Tree (LMT) algorithm, and the J48 algorithm (Quinlan, 1986). The RF algorithm works with multiple internal decision trees to obtain final classification results through its internal classification results combined (Ho, 1995; Svetnik et al., 2003). The LMT algorithm combines Logistic Regression and decision tree methods (Landwehr et al., 2005). The J48 algorithm is a C4.5 algorithm Java implementation that works with decision trees using the information entropy concept (Quinlan, 2014). RF and J48 are included since they provide good classifying texts (Gabrilovich & Markovitch, 2004; Fersini, Pozzi & Messina, 2015). LMT is also included since it is one of the best algorithms to classify texts using the stylistic resources of the Portuguese language (Aires et al., 2004).

The WEKA software returns the values of precision (P), recall (R), and the F-measure for each of the algorithms using the k-fold cross-validation technique (Kohavi, 1995). F-measure is a harmonic mean of P and R. F-measure ranges between ‘0’ and ‘1’, where ‘0’ indicates the worst value and ‘1’ indicates the best value, i.e., perfect values of P and R.

## 5 Experimental evaluation

After obtaining the datasets, each publication is separated into a single file, allotted in a respective class folder to facilitate the generation of the ARFF format used in WEKA. No additional pre-processing is required, as the LIWC program does not distinguish between uppercase and lowercase spelling. All files were analyzed in WEKA to support the experimental evaluation of this work. As stated in the introduction, this work investigates three questions: (i) Does BP-LIWC2015 achieves better text classification results than the older Brazilian Portuguese dictionary (BP-LIWC2007)? (ii) What is the equivalence between BP-LIWC2015 and BP-LIWC2007 with LIWC2015? (iii) Are there significant differences between Brazilian Portuguese dictionaries? Section 5.1 presents a set of classification experiments with four datasets and seven classification algorithms to explore the first question. The second is examined with correlation experiments in Sect. 5.2. The last question is considered in Sect. 5.3.

**Table 3** F-measure for classification algorithms to infer MQD users age range, using all 73 BP-LIWC2015 categories and 64 BP-LIWC2007 categories

| | NB | MNB | J48 | RF | LMT | SMO | LR |
|---|---|---|---|---|---|---|---|
| BP-LIWC2007 | 0.536 | 0.527 | 0.521 | 0.566 | 0.563 | 0.440 | 0.530 |
| BP-LIWC2015 | **0.538** | **0.543** | **0.528** | **0.572** | **0.570** | **0.458** | **0.535** |

**Table 4** F-measure for classification algorithms to define MQD users' polarities of emotions

| | NB | NBM | J48 | RF | LMT | SMO | LR |
|---|---|---|---|---|---|---|---|
| BP-LIWC2007 | 0.631 | 0.711 | 0.682 | 0.752 | 0.753 | 0.753 | 0.753 |
| BP-LIWC2015 | **0.637** | **0.725** | **0.710** | **0.777** | **0.775** | **0.773** | **0.775** |

**Table 5** F-measure for classification algorithms to define Twitter users polarities of emotions (TAS-PT-60 K).

| | NB | MNB | J48 | RF | LMT | SMO | LR |
|---|---|---|---|---|---|---|---|
| BP-LIWC2007 | 0.615 | 0.649 | 0.644 | 0.697 | 0.683 | 0.668 | 0.673 |
| BP-LIWC2015 | **0.743** | **0.875** | **0.949** | **0.955** | **0.965** | **0.953** | **0.959** |

### 5.1 Classification experiments

More meaningful features result in greater accuracy in the classification task, as described in Sect. 1. We adopted this principle to examine the quality of features extracted by BP-LIWC2015 and BP-LIWC2007. Thus, we processed data from four datasets with both dictionaries. Next, we adopted seven classification algorithms to evaluate the performance of the classification algorithms using the data provided by these dictionaries.

The first experiment inferred MQD users' age range in the MQD90k dataset. The classification was conducted using all 73 BP-LIWC2015 categories and 64 BP-LIWC2007 categories. Table 3 presents the F-measure results provided by the classification algorithms (i.e., NB, MNB, J48, RF, LMT, SMO, and LR). All the algorithms obtained the best results using BP-LIWC2015. In addition, the RF algorithm produced better results than the other algorithms (i.e., 0.572).

The second classification experiment was performed on the MQD60k dataset. This experiment processed MQD60k with all 73 BP-LIWC2015 categories and all 64 BP-LIWC2007 categories to perform a classification task in MQD users' polarities of emotions (i.e., happiness and sadness). Table 4 shows the F-measure results for each classification algorithm. The results using BP-LIWC2015 outperform BP-LIWC2007 with all algorithms. RF also presented better results than other algorithms (i.e., 0.777). When considering only BP-LIWC2007 results LMT and LR achieved more promising results, but the performance still increased by 2.92% using BP-LIWC2015.

The third experiment was conducted with the TAS-PT-60 K dataset, where the Portuguese Twitter posts were also processed using both the 73 BP-LIWC2015 and 64 BP-LIWC2007 categories. Table 5 presents F-measure results acquired with NB, MNB, J48, RF, LMT, SMO, and LR to infer Twitter users' polarities of emotions (i.e., positive and negative). BP-LIWC2015 outperforms BP-LIWC2007 with all classifi-

cation algorithms. The LMT and the LR algorithms achieve the highest F-measure with BP-LIWC2015.

The last classification experiment was performed using the TW-PTSA-100k dataset. This experiment also considers classification tasks in Portuguese Twitter posts. The results were also compared using the 73 BP-LIWC2015 and 64 BP-LIWC2007 categories. Table 6 brings the results provided by NB, MNB, J48, RF, LMT, SMO, and LR. This classification was also conducted to infer Twitter users' polarities of emotions (i.e., positive and negative). One can note that all algorithms achieved better results with BP-LIWC2015. The LMT algorithm obtained the best results (i.e., 0.940).

The results presented in Tables 3, 4, 5 and 6 were tested for normality using the Kolmogorov–Smirnov test before performing the paired T-test, with a confidence level of 99% and degree of freedom DF = 6. Tables 3, 4 and 5, and 6 respectively present p-values of 0.008, 0.0004, 0.00003, and 0.00006, indicating a statistically significant difference between the dictionaries. These results indicate with more than 99% confidence that the usage of BP-LIWC2015 resulted in better F-measure for all classification algorithms.

### 5.2 Correlation experiments

Eichstaedt et al. (2021) reported that the 2015 English dictionary (i.e., LIWC2015) clearly improves upon the 2007 English dictionary (LIWC2007). Since the classification experiments indicated BP-LIWC2015 outperformed BP-LIWC2007, we expect that BP-LIWC2015 presents stronger correlation with LIWC2015 than BP-LIWC2007. In line with this reasoning, this section describes a set of correlation experiments to assess the equivalence between LIWC2015, BP-LIWC2015, and BP-LIWC2007. Correlation experiments aim to test the equivalence between different dictionaries as a measure of criterion validity, that is, the capacity of two dictionaries to capture the same trends within similar datasets (Dudău & Sava, 2020).

In this scenario, this section first introduces correlation experiments between BP-LIWC2015 and BP-LIWC2007, adopting Kendall's $\tau_B$ correlation coefficient. Next, it presents correlation experiments between LIWC2015 and BP-LIWC2007 in contrast to LIWC2015 and BP-LIWC2015, also adopting Kendall's $\tau_B$. Kendall's $\tau_B$ assesses statistical associations and does not depend on assumptions about the distributions (Noether, 1981). This approach is grounded on three literature studies: Wissen and Boot (Wissen & Boot, 2017), Meier et al. (Meier et al., 2019), and Dudău and Sava (Dudău & Sava, 2020).

To begin, we address the criterion validity of BP-LIWC2015 within the Brazilian Portuguese language. This strategy aims to assess the equivalence of BP-LIWC2015 with the older translation (i.e., BP-LIWC2007) since correlations values between

**Table 6** F-measure for classification algorithms to define Twitter users polarities of emotions (TW-PTSA-100k).

| | NB | NBM | J48 | RF | LMT | SMO | LR |
|---|---|---|---|---|---|---|---|
| BP-LIWC2007 | 0.613 | 0.646 | 0.655 | 0.707 | 0.695 | 0.672 | 0.676 |
| BP-LIWC2015 | **0.716** | **0.835** | **0.921** | **0.933** | **0.940** | **0.933** | **0.934** |

word categories can reflect a general equivalence between the two dictionaries. Thus, in the first step, each text of the bilingual Portuguese-English corpus FAPESP-CORPUS was processed using LIWC2015, BP-LIWC2015, and BP-LIWC2007, producing a table similar to Table 1 (illustrated in Sect. 3).

Table 7 presents the equivalence between the BP-LIWC2007 and BP-LIWC2015 regarding FAPESP-CORPUS. The first column describes the main category names; the second and third columns show median values provided by the BP-LIWC2015 and the BP-LIWC2007, respectively. Finally, the fourth column shows the equivalence between the two versions of the BP dictionaries, assessed with Kendall's $\tau_B$ correlation coefficient.

Finally, we calculated the equivalence between LIWC2015 and BP-LIWC2007 ($\tau_{B1}$) and the equivalence between LIWC2015 and BP-LIWC2015 ($\tau_{B2}$). The results are presented in Table 8. It is possible to note that the equivalence between LIWC2015 and BP-LIWC2015 shows higher Kendall's $\tau_B$ correlation coefficients. The 'Adjectives', 'Comparisons', 'Interrogatives', 'Drives' and 'Informal Language' categories do not have an equivalent match in BP-LIWC2007, and thus no correlation value is available. Since the FAPESP-CORPUS consists of scientific texts, only a small percentage of words in the 'Informal Language' category can be detected, hindering results for this category.

### 5.3 Coverage experiments

BP-LIWC2007 and BP-LIWC2015 have 127,149 and 14,459 items in their dictionaries, respectively. The notable difference in the number of items between the

**Table 7** Equivalence between the BP-LIWC2007 and BP-LIWC2015: median and Kendall's $\tau_B$ correlation coefficients in the values of the main categories

| Category | BP-LIWC2007 | BP-LIWC2015 | $\tau_B$ |
|---|---|---|---|
| Function words | 46.43 | 46.81 | 0.670 |
| Verbs | 10.29 | 7.92 | 0.478 |
| Numbers | 3.36 | 4.55 | 0.784 |
| Quantifiers | 4.46 | 2.01 | 0.414 |
| Affect | 4.93 | 2.60 | 0.479 |
| Social | 10.64 | 3.93 | 0.385 |
| Cognitive Processes | 17.20 | 10.09 | 0.472 |
| Perceptual Processes | 2.44 | 1.83 | 0.290 |
| Biological Processes | 3.43 | 1.64 | 0.533 |
| Past Focus | 2.54 | 1.38 | 0.608 |
| Present Focus | 6.64 | 5.22 | 0.656 |
| Future Focus | 0.48 | 0.32 | 0.191 |
| Relativity | 21.34 | 12.45 | 0.528 |
| Work | 5.60 | 4.23 | 0.718 |
| Leisure | 1.22 | 0.41 | 0.368 |
| Home | 0.30 | 0.08 | 0.373 |
| Money | 2.16 | 0.71 | 0.506 |
| Religion | 0.32 | 0.06 | 0.409 |
| Death | 0.16 | 0.00 | 0.366 |

**Table 8** Results of the analysis of FAPESP-CORPUS for equivalence between LIWC2015 x BP-LIWC2007 ($\tau_{B1}$) and LIWC2015 x BP-LIWC2015 ($\tau_{B2}$), in terms of Kendall's $\tau_B$ correlation coefficient

| Category | $\tau_{B1}$ | $\tau_{B2}$ |
|---|---|---|
| Function words | 0.46 | **0.54** |
| Verbs | **0.48** | 0.46 |
| Adjectives | – | **0.47** |
| Comparisons | – | **0.56** |
| Interrogatives | – | **0.43** |
| Numbers | **0.73** | 0.68 |
| Quantifiers | 0.33 | **0.55** |
| Affect | 0.46 | **0.59** |
| Social | 0.37 | **0.60** |
| Cognitive Processes | 0.44 | **0.60** |
| Perceptual Processes | **0.41** | 0.40 |
| Biological Processes | 0.47 | **0.64** |
| Drives | – | **0.59** |
| Past Focus | 0.56 | **0.63** |
| Present Focus | **0.54** | 0.52 |
| Future Focus | 0.18 | **0.48** |
| Relativity | 0.45 | **0.57** |
| Work | 0.73 | **0.77** |
| Leisure | 0.33 | **0.44** |
| Home | 0.35 | **0.49** |
| Money | 0.50 | **0.60** |
| Religion | 0.36 | **0.38** |
| Death | 0.41 | **0.72** |
| Informal Language | – | **0.01** |

two dictionaries draws attention to the better results achieved with BP-LIWC2015 in classification experiments. From this perspective, it is noteworthy to investigate the coverage of both dictionaries. To accomplish this goal, we conducted coverage experiments with the bilingual Portuguese-English corpus FAPESP-CORPUS.

We processed all FAPESP-CORPUS text files with LIWC2015, BP-LIWC2015, and BP-LIWC2007. Table 9 shows, for each main category, the median[1] min, and max values to indicate the coverage of LIWC2015, BP-LIWC2007, and BP-LIWC2015 concerning FAPESP-CORPUS[2]. Values are expressed as the percentage of total words used in any given language sample.

Results reveal BP-LIWC2015 covers more words than BP-LIWC2007 in seven categories: *function words*, *adjectives*, *comparisons*, *interrogatives*, *numbers*, *drives*, and *informal language*. In all other 17 categories, BP-LIWC2007 covered more words than BP-LIWC2015. It is appropriate to point out that BP-LIWC2015 and LIWC2015 coverage medians are more closely related than BP-LIWC2007 and LIWC2015; in

[1] The median was chosen as the dispersion measure since the values found for most categories corresponded to a non-normal distribution according to the D'Agostino-Pearson test (D'Agostino & Belanger, 1990).

[2] To clarify the procedure conducted, revisit Table 1. After the n texts have been processed, each column's max, min, and median values (i.e., category) are found. Thus, considering the five values presented in Table 1, the Function words category has a median equal to 21.73.

**Table 9** Median, minimum (min), and maximum (max) values of the main categories to indicate the coverage of LIWC2015, BP-LIWC2007, and BP-LIWC2015 with reference to FAPESP-CORPUS.

| | LIWC2015 | | | BP-LIWC2007 | | | BP-LIWC2015 | | |
|---|---|---|---|---|---|---|---|---|---|
| Category | Median | Min | Max | Median | Min | Max | Median | Min | Max |
| Function words | 46.90 | 9.52 | 57.75 | 46.43 | 10.07 | 58.20 | 46.81 | 10.74 | 58.25 |
| Verbs | 9.00 | 0.00 | 17.23 | 10.29 | 1.34 | 17.60 | 7.92 | 0.00 | 17.16 |
| Adjectives | 3.95 | 0.00 | 9.77 | NA | NA | NA | 3.91 | 1.20 | 8.89 |
| Comparisons | 2.14 | 0.00 | 6.64 | NA | NA | NA | 2.53 | 0.00 | 6.02 |
| Interrogatives | 1.19 | 0.00 | 5.69 | NA | NA | NA | 3.79 | 0.00 | 11.00 |
| Numbers | 2.58 | 0.00 | 58.88 | 3.36 | 0.00 | 56.67 | 4.55 | 0.00 | 56.67 |
| Quantifiers | 1.74 | 0.00 | 5.47 | 4.46 | 0.00 | 9.25 | 2.01 | 0.00 | 6.76 |
| Affect | 2.54 | 0.00 | 9.27 | 4.93 | 0.00 | 15.27 | 2.60 | 0.00 | 9.79 |
| Social | 4.26 | 0.00 | 15.34 | 10.64 | 0.00 | 27.00 | 3.93 | 0.00 | 13.06 |
| Cognitive Processes | 8.21 | 0.93 | 17.49 | 17.20 | 0.93 | 48.35 | 10.09 | 0.47 | 21.62 |
| Perceptual Processes | 1.24 | 0.00 | 11.02 | 2.44 | 0.00 | 11.02 | 1.83 | 0.00 | 8.85 |
| Biological Processes | 1.10 | 0.00 | 11.32 | 3.43 | 0.00 | 16.40 | 1.64 | 0.00 | 10.77 |
| Drives | 5.66 | 0.00 | 14.84 | NA | NA | NA | 7.09 | 0.00 | 15.76 |
| Past Focus | 2.72 | 0.00 | 9.80 | 2.54 | 0.00 | 9.70 | 1.38 | 0.00 | 8.21 |
| Present Focus | 5.55 | 0.00 | 12.78 | 6.64 | 1.11 | 12.31 | 5.22 | 0.00 | 10.47 |
| Future Focus | 0.54 | 0.00 | 4.39 | 0.48 | 0.00 | 2.99 | 0.32 | 0.00 | 3.96 |
| Relativity | 12.88 | 0.79 | 24.85 | 21.34 | 8.89 | 34.63 | 12.45 | 3.29 | 25.35 |
| Work | 5.36 | 0.47 | 20.42 | 5.60 | 0.69 | 18.69 | 4.23 | 0.00 | 16.19 |
| Leisure | 0.38 | 0.00 | 8.24 | 1.22 | 0.00 | 8.89 | 0.41 | 0.00 | 8.70 |
| Home | 0.08 | 0.00 | 4.97 | 0.30 | 0.00 | 5.23 | 0.08 | 0.00 | 4.58 |
| Money | 0.50 | 0.00 | 9.12 | 2.16 | 0.00 | 14.58 | 0.71 | 0.00 | 10.76 |
| Religion | 0.03 | 0.00 | 7.24 | 0.32 | 0.00 | 7.86 | 0.06 | 0.00 | 6.32 |
| Death | 0.00 | 0.00 | 3.24 | 0.16 | 0.00 | 3.45 | 0.00 | 0.00 | 3.26 |
| Informal Language | 0.20 | 0.00 | 2.79 | NA | NA | NA | 0.25 | 0.00 | 1.97 |

some cases (e.g., as the *home* category), LIWC2015 and BP-LIWC2015 present the same coverage.

### 5.4 Discussion

The first set of experiments (Sect. 5.1) was conducted to evaluate BP-LIWC2015 and BP-LIWC2007 with four Brazilian Portuguese datasets from real social networks. Two datasets were obtained from MQD social network, and the other two were extracted from Twitter. The results were acquired by running seven machine learning classification algorithms and analyzed using the F-measure.

The BP-LIWC2015 outperformed BP-LIWC2007 in all datasets with all classification algorithms. In the datasets from MQD, the best results were obtained by RF. On the other hand, the best results acquired with Twitter datasets were performed by LTM. These results are supported by the literature, which states that RF provides good results in classifying texts, and LTM is one of the best algorithms to classify texts using the stylistic resources of the Portuguese language. It is necessary to highlight the F-measure scores difference observed in Twitter datasets considering BP-LIWC2015 and BP-LIWC2007. This behavior can be explained by the Twitter post's length, which, unlike MQD, has character limitations.

The second set of experiments (Sect. 5.2) was designed to understand the better results of BP-LIWC2015 in the classification experiments. The goal was first to calculate the correlation between BP-LIWC2015 and BP-LIWC2007 to point out their equivalence. From the 19 main categories, eight presented a moderate correlation (i.e., between 0.4 and 0.6), six showed a weak correlation (i.e., between 0.1 and 0.39), and five achieved a strong correlation (between 0.6 and 0.9). No perfect correlation was found. Meier et al. (2019) describe that the lower correlation indicates more changes across the two versions. Following this line of thought, the results reveal a meaningful difference between BP-LIWC2015 and BP-LIWC2007.

Next, Sect. 5.2 presented the equivalence between LIWC2015 and BP-LIWC2015 in addition to the equivalence between BP-LIWC2007 and BP-LIWC2015. The goal was to investigate which one would lead to a stronger correlation with LIWC2015. Although BP-LIWC2007 contains more words than BP-LIWC2015 (127,149 and 14,459, respectively), it does not imply a stronger correlation between BP-LIWC2007 and LIWC2015. One can note that by analyzing the 20 main categories, BP-LIWC2015 presented a more substantial correlation with LIWC2015 in 16 categories, indicating that, in line with our initial expectation, BP-LIWC2015 has a better equivalence with LIWC2015 than BP-LIWC2007.

The literature reported that LIWC2015 improves upon LIWC2007. On the other hand, this study indicated that: (i) BP-LIWC2015 outperformed BP-LIWC2007 in classification experiments; (ii) BP-LIWC2015 exhibited a stronger correlation with LIWC2015; and (iii) BP-LIWC2015 showed a low correlation with BP-LIWC2007. These findings are consistent since BP-LIWC2015 is based on LIWC2015 and BP-LIWC2007 on LIWC2007. Thus, the better results acquired by BP-LIWC2015 compared to BP-LIWC2007 can be justified by a higher correlation with LIWC2015.

Finally, this study investigated the coverage of BP-LIWC2015, BP-LIWC2007, and LIWC2015. The results indicated that BP-LIWC2015 covers more words than BP-LIWC2007 in 26% (i.e., seven categories) of the categories and fewer words in 74% (i.e., 17 categories). It is also important to note that, although BP-LIWC2015 covers more words in seven categories, five do not exist in BP-LIWC2007; they are new in BP-LIWC2015. These categories are *adjectives*, *comparisons*, *interrogatives*, *numbers*, *drives*, and *informal language*. Despite having a lower coverage, BP-LIWC2015 achieved better results in classification experiments. Two main reasons can justify this. The first refers to the fact that a higher coverage does not indicate a better categorization of items in the dictionaries. Problems related to BP-LIWC2007 include items with missing categories or improperly assigned, as discussed in Sect. 3. The second refers to the new categories in BP-LIWC2015. The improved classification results with BP-LIWC2015 can also be linked to the new categories since data features were generated based on dictionary categories. In this regard, Fukunaga (1990) states that better class separability depends on the data features and the classifier used. The author also indicates that class separability can boost classification performance, possibly explaining the better results achieved with BP-LIWC2015.

## 6 Conclusion

LIWC is one of the vastly used and most powerful computer-based language analysis tool. For example, it can be applied to verify threats to social and economic conditions, detect mental illnesses, assert marketing tendencies, and identify deception. Researchers worldwide have contributed over the years to translate the LIWC dictionaries into languages such as English, Spanish, Dutch, German, Romanian, and Brazilian Portuguese. The latter was recently released and needed further evaluation, as reported in the literature.

In this scenario, this work evaluated the Brazilian Portuguese 2015 LIWC dictionary using classification algorithms and texts from social networks. It also includes a statistical comparison of both Brazilian Portuguese LIWC dictionaries (i.e., 2007 and 2015) to measure the quality of each main category's content. The methodology used in this study followed the literature that evaluates LIWC dictionaries in other languages.

This study uses publicly available datasets and employs different machine learning classification algorithms to enable reproducibility. The results indicate that BP-LIWC2015 outperforms BP-LIWC2007 in the classification task. Moreover, the results showed a greater equivalence between LIWC2015 and BP-LIWC2015 than between LIWC2015 and BP-LIWC2007.

For future work, we intend to perform psychometric evaluations for the categories and subcategories of LIWC dictionaries. This evaluation may be helpful to revise words in categories and for meaningful improvement in dictionary coverage. We also intend to perform more diverse classification tasks, e.g., education level or political homophily, to explore the dictionary's versatility and usefulness for other research areas.

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